Goal 2: A Frequent Relation Is Learned as Nonlinear

factual-recall-mechanisms

2026-07-03

1 Claim and result in one paragraph

A small transformer is trained for **exactly one epoch** on a synthetic corpus containing two relations with *identical frequency statistics at every level* (same subjects, same 32 candidate objects, same number of triples, same number of occurrences per triple): a semantically supported relation R_{sem} and a random relation R_{rnd}. At *high* fact frequency (F=64 occurrences per triple) both relations are *fully learned*: recall is 1.000 on all 128 training triples for both, in all three seeds. Yet an affine map fitted by ordinary least-squares linear regression from subject representations to object representations identifies held-out targets at 0.891 ± 0.056 for R_{sem} and at 0.016 ± 0.022 (= chance) for R_{rnd}. The frequent random relation is therefore stored, but stored *non-linearly*: no affine map over the same representations accounts for it, and it does not generalize (held-out recall at chance). Because the semantic relation at the very same frequency remains linear, the nonlinearity is attributable to the relation's semantics, not to its frequency; conversely, raising frequency from 1 to 64 occurrences never makes the random relation linear.

2 Data

Vocabulary (265 tokens). A separator `x`; two relation tokens `r_0`, `r_1`; 192 subject tokens `a_0`, ..., `a_191`; 32 object tokens `b_0`, ..., `b_31`; 32 attribute tokens `f_0`, ..., `f_31`; 6 attribute tokens `g_0`, ..., `g_5`.

Entities and attributes. Subject `a_i` carries the attribute pair $(f_{i \bmod 32},\ g_{\lfloor i/32 \rfloor})$, so the 192 subjects realize each of the 32×6 attribute combinations exactly once. Object `b_j` carries $(f_{(j+3) \bmod 32},\ g_0)$.

Sentences. All training sentences have five tokens:

attribute sentence	`x a_i f_p g_q x`	and	`x b_j f_p g_q x`
relation triple	`x a_i r_r b_t x`		

Relations.

$$R_{\text{sem}}(a_i) = b_{f(i)}, \qquad R_{\text{rnd}}(a_i) = b_{\rho(i)},$$

where $f(i) = i \bmod 32$ is the subject's own f-attribute index — so the semantic target is the object whose f-attribute is the subject's, shifted by 3 — and ρ is a seed-dependent balanced random assignment onto the same 32 objects, independent of the attributes. Concrete example (seed 19): `a_37` has attributes (`f_5`, `g_1`), so R_{sem}(`a_37`) = `b_5`, while R_{rnd}(`a_37`) = `b_27`; the training sentences are `x a_37 r_0 b_5 x` and `x a_37 r_1 b_27 x`.

Split and exact frequency matching. For every `f_`-value, 4 of its 6 subjects are *trained* and 2 are *held out* (128 trained, 64 held-out subjects; seed-dependent). ρ is constructed balanced within both groups. Consequently, under *both* relations: every trained subject occurs in exactly one triple per relation; every object is the target of exactly 4 trained and 2 held-out subjects; both relations contribute exactly 128 training triples, each repeated the same number of times (F). The two relations are indistinguishable by any frequency statistic; only the assignment semantics differ. Held-out subjects appear in attribute sentences but never in any relation triple. These invariants are asserted by `tests/test_linear_shift.py`.

3 Model and training

Model	causal transformer, 1 layer, hidden 64, 4 heads, context 8
Embeddings	input and output embeddings untied; all weights $\mathcal{N}(0, 0.02^2)$
	no attribute-based initialization; nothing frozen; no loss reweighting
Parameters	84,288
Corpus	every attribute sentence $\times A$=128 (= 28,672 sentences)
	+ 128 triples per relation, each $\times F$ ($F \in \{1, 4, 16, 64\}$)
Training	**exactly one epoch**: a single pass over the shuffled corpus
Steps	452 (F=1) to 704 (F=64) steps at batch size 64
Objective	unweighted next-token cross-entropy on all positions
Optimizer	AdamW, constant LR 10^{-3} after 5 linear warmup steps,
	weight decay 0, gradient clip 1.0
Seeds	19, 31, 47 (control data assignment, initialization, and data order)

Note that "frequent" here means frequent within a single epoch: at F=64 the corpus contains each triple 64 times, but the model still passes over the corpus exactly once.

4 The metric

A single analysis metric is used, as an operationalization of "the relation is an affine transformation":

> Fit $\hat{W}, \hat{b}$ by ordinary least-squares linear regression from the subject's *input embedding* $s_i \in \mathbb{R}^{64}$ to the target object's *output embedding* (unembedding row) $o_{R(i)} \in \mathbb{R}^{64}$, using the 128 trained pairs. For each of the 64 held-out subjects, predict $\hat{W}s_j + \hat{b}$ and select the nearest object row by cosine similarity among all 32 objects. Report the fraction of held-out subjects for which the selected object is $R(j)$. Chance is $1/32 \approx 0.03$.

For every subject exactly one object is correct under each relation, so no top-k metrics are needed or used. In addition, one *learning check* (not an analysis metric) is recorded: greedy next-token recall of the model on prompts `x a_i r_r`, for trained and held-out subjects. For the claim "learned as nonlinear" this check is essential: it establishes that the relation *is* stored.

5 Results

Finding 1: a fully learned frequent relation with no linear structure (F=64). At 64 occurrences per triple, R_{rnd} reaches perfect recall of all 128 training triples — the relation is fully stored — while the affine probe remains at chance (0.016 ± 0.022) and held-out recall is at chance (.031). The storage is a non-linear, non-generalizing memorization.

	probe accuracy (held-out)		recall (trained)		recall (held-out)	
F	R_{sem}	R_{rnd}	R_{sem}	R_{rnd}	R_{sem}	R_{rnd}
1	$1.000 \pm .000$	$.031 \pm .013$	.029	.010	.016	.010
4	$1.000 \pm .000$	$.016 \pm .013$	.055	.034	.057	.031
16	$0.990 \pm .007$	$.036 \pm .007$	.672	.167	.635	.031
64	$0.891 \pm .056$	$\mathbf{.016 \pm .022}$	**1.000**	**1.000**	.995	**.031**

Table 1: Mean $\pm$ sd over seeds 19/31/47, at A=128 attribute-sentence occurrences. F is the number of occurrences of each of the 128 triples per relation. Recall columns are the learning check. The headline cell is F=64: both relations are fully stored (recall 1.000), but only R_{sem} admits an affine map.

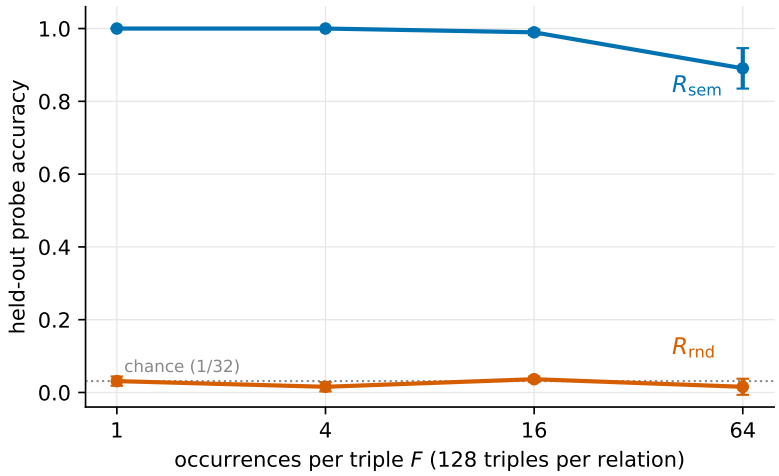


Figure 1: Held-out probe accuracy vs. occurrences per triple, for the two frequency-matched relations. The random relation is never linear, at any frequency; the semantic relation is linear at every frequency, with slight erosion at F=64.

Finding 2: the nonlinearity is due to semantics, not frequency. R_{sem}, trained in the same model from data with *identical* frequency statistics, is equally fully stored (recall 1.000) but linear (probe 0.891 ± 0.056) and generalizing (held-out recall .995). Conversely, increasing the frequency of R_{rnd} from 1 to 64 occurrences never produces linear structure (Fig. 1): frequency determines only *whether* the random relation gets stored (recall .010 $\rightarrow$ 1.000), never *in what format*.

Finding 3: high frequency slightly erodes the linear format of the semantic relation. R_{sem}'s probe declines from 1.000 ($F \leq 4$) to $0.891 \pm .056$ (F=64), consistent with high-frequency memorization beginning to compete with the linear solution, while remaining far from the random relation's chance level.

6 Mechanistic forward-pass analysis

The failed held-out affine probe establishes an operationally nonlinear, non-generalizing storage format, but does not identify the computation that recalls the 128 trained random facts. We

therefore reconstructed every F=64 checkpoint by deterministic replay, requiring final loss, probe accuracy, trained recall, and held-out recall to match the archived JSON to $< 10^{-7}$. For each seed, the 128 trained facts were split into target-stratified discovery and evaluation halves. All head/neuron selection used the discovery half; every number below is computed on the disjoint evaluation half.

At the final prompt position of `x a_i r_1`, the model computes four attention-head outputs, adds their projected sum to the relation residual, applies 256 GELU neurons position-wise, and unembeds the final normalized residual. A logit lens localizes the recall computation: object accuracy is $.031 \pm .000$ at the bare `r_1` residual, $.927 \pm .052$ immediately after attention, and $1.000 \pm .000$ after the MLP.

The used circuit. Mean-ablating all four heads drops object accuracy from $1.000 \pm .000$ to $.031 \pm .000$. Exhaustive evaluation of all 2^4 head subsets and exact margin Shapley values identify H1 as the stable primary lookup head, H0/H2 as seed-variable supporting heads, and H3 as causally negligible. In clean/corrupt subject swaps, patching all head outputs restores $1.000 \pm .000$ of the clean-target logit effect. Patching only the value vectors restores $.975 \pm .020$, while patching only the attention patterns restores $.023 \pm .014$. Thus the arbitrary fact is encoded in the subject-dependent value content, not in a subject-specific routing pattern: swapping the transported value code moves the output toward the other subject's arbitrary target.

The post-attention state already recalls most facts, and a distributed MLP sharpens the result. Mean-ablating the 32 highest-ranked neurons leaves $.826 \pm .014$ of the subject-specific MLP margin, while keeping only those 32 variable neurons recovers $.478 \pm .022$. There is no universal single fact neuron, and physical neuron IDs vary by initialization. Holding the subject fixed and swapping only `r_1` $\leftrightarrow$ `r_0` shows that relation selection is present in both stages: patching the four random-relation head outputs restores $.922 \pm .024$ of the relation effect, and patching the MLP population restores $.841 \pm .032$.

Crucially, "nonlinear storage" must not be strengthened into an unsupported claim that GELU curvature itself performs the lookup. A per-neuron affine surrogate fitted to the GELUs on discovery subjects attains $1.000 \pm .000$ object accuracy on the disjoint memorized facts (activation $R^2 = .924 \pm .006$); deleting the MLP still leaves $.927 \pm .052$. Most trained-fact lookup therefore resides in learned subject keys and their attention value paths, with the MLP adding margin and correcting residual errors. The storage is called nonlinear in the precise sense tested in this paper: an affine map fitted from trained subject embeddings to object rows is at chance on held-out subjects, and the behavior itself fails to generalize. It is not a claim that the finite trained lookup cannot be locally approximated by affine operations.

The frequent random relation is therefore recalled by a non-generalizing key–value memory: relation-gated heads retrieve a subject-specific value code, and a distributed MLP/readout sharpens it into the memorized object direction. There is no shared rule for subjects outside the memorized set, explaining the simultaneous perfect training recall, chance held-out recall, and chance held-out affine probe.

7 What this does and does not show

- **Shown:** a frequent relation can be fully learned and stored non-linearly (no affine map over the representations accounts for it, and it does not generalize); with frequency exactly matched, storage format is decided by semantics.

- **Qualification:** "frequent ⇒ nonlinear" holds for the relation without attribute support. Frequency alone does not force nonlinearity: the frequent *semantic* relation remains largely linear (with mild erosion at F=64). The precise claim supported is: at matched high frequency, an attribute-supported relation is stored linearly and an unsupported one non-linearly.
- **Superseded:** earlier drafts' high-frequency results used attribute-initialized embeddings (`structured_embedding_init`), targeted-row updates, and top-k metrics; they are replaced by the present end-to-end setup with a single OLS metric.

8 Reproduction

Implementation: `experiments/factual_recall/linear_shift.py` (self-contained). Design invariants: `experiments/tests/test_linear_shift.py`. From `experiments/`:

```
# Table 1 / Figure: one seed per command (19, 31, 47), ~1 min each on one GPU
uv run python -m factual_recall.linear_shift --world category --probe embedding \
  --seeds 19 --k-values 0,128 --occurrences 1,4,16,64 --attr-repeats 128 \
  --device cuda --output-dir factual_recall/results/final_category_seed19

# Aggregation/figures: factual_recall/results/final_summary/
uv run python -m factual_recall.linear_shift_mechanistic_analysis --device cuda
uv run pytest tests/test_linear_shift.py -q
```

Raw per-run metrics (including probe R^2/cosine and losses) are in `factual_recall/results/final_category_`
The companion document `goal1_infrequent_relation_linear.pdf` covers the infrequent/linear claim on the same runs.

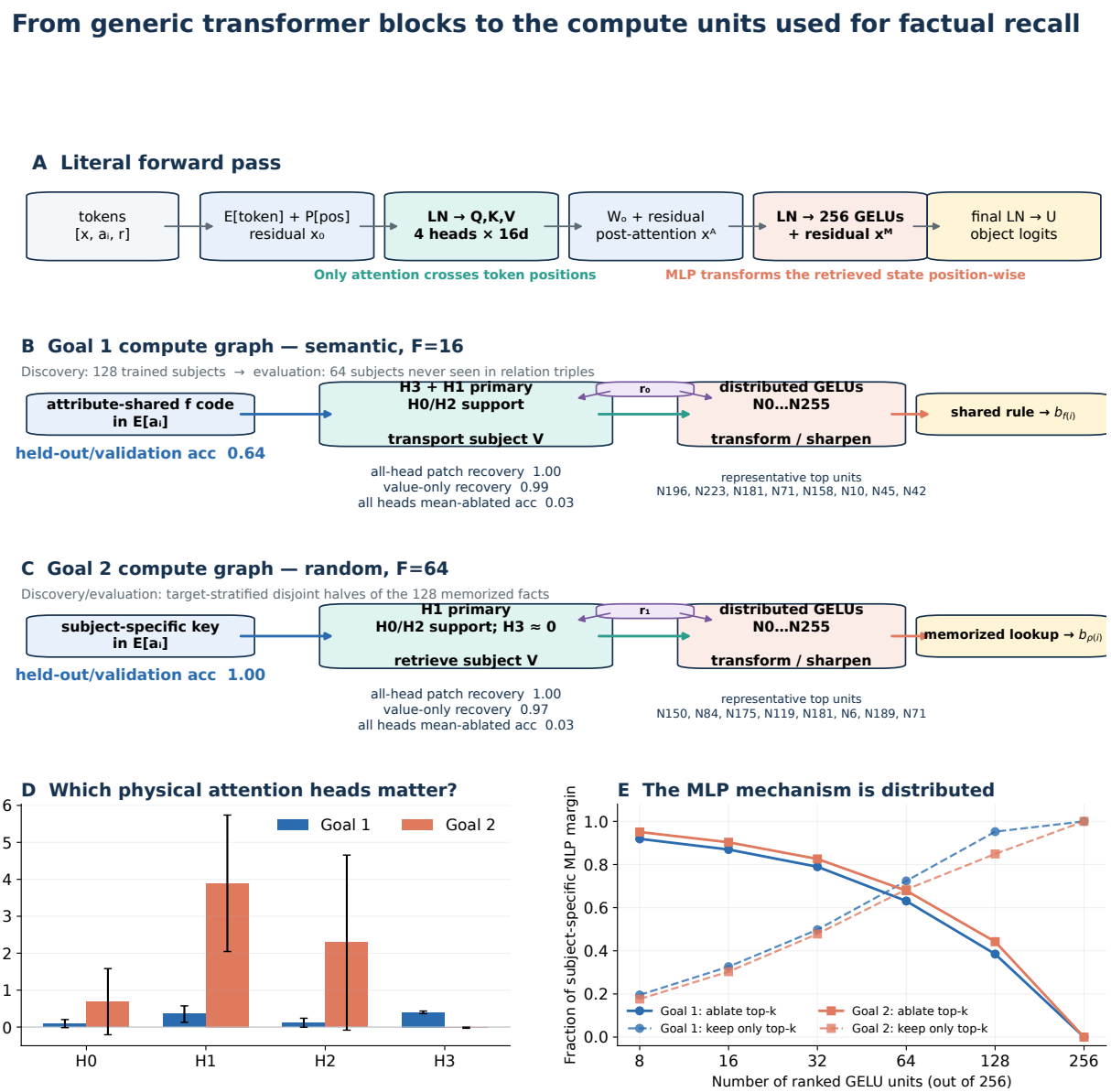


Figure 2: Top: the literal forward pass. Middle: generic attention and MLP blocks replaced by the causally identified Goal 1 and Goal 2 compute units. Bottom: exact head Shapley effects and discovery/evaluation-split neuron concentration curves, aggregated over seeds 19/31/47.